

Day 22 Plan

Formula sheet · reasoning rules · mistake log · GA fact list · exam-day checklist
· sleep by 10:30 pm

Today's schedule (about 3½ hours — light day)

#	Block	Time	What you do
1	Maths formula sheet	60 min	Read the sheet below; for each line, solve one quick example in your head
–	Break	10 min	Walk
2	Mistake log	45 min	Read every mistake from Days 1-21; write your personal top-10 traps on one card
3	Science	45 min	Day 21's S-tier sheet once more, covering the right column
4	GA fact list	30 min	The list below + the last 2 months of current affairs headlines
5	Logistics	15 min	Checklist at the end: admit card, ID, route, reporting time

No full mock and no new topics today. If your exam is on 28 Oct, repeat this light day on 27 Oct (plus one 25-question sectional).

Maths formula sheet

Topic	Formulas
Percentage	Successive a%, b%: $a + b + \frac{ab}{100}$ · A is r% more than B $\rightarrow$ B is $\frac{r}{(100 + r)} \times 100\%$ less · population $P(1 + \frac{r}{100})^n$
Ratio	$a : b$ and $b : c \rightarrow$ multiply to match b · partnership profit $\propto$ capital $\times$ time
Profit, loss	Profit% on CP · SP = $CP(1 + \frac{p}{100})$ · successive discounts $a + b - \frac{ab}{100}$ · false weight gain = $\frac{\text{error}}{(\text{true} - \text{error})} \times 100$
SI, CI	SI = $\frac{PRT}{100}$ · A = $P(1 + \frac{R}{100})^n$ · CI – SI (2 yrs) = $P(\frac{R}{100})^2$
Number system	Divisibility 3, 9 digit sum; 4 last two digits; 8 last three; 11 alternate-sum difference · unit digit cycle of 4 · LCM $\times$ HCF = product of two numbers
Average	Sum = avg $\times$ n · replacement: new = old + n $\times$ rise · joining: new = avg + (n + 1) $\times$ rise · $\frac{2xy}{(x + y)}$
Time & work	Total = LCM · efficiency = total/days · $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$ · leak negative · wages $\propto$ work
TSD, boats	$\times \frac{5}{18}$ · trains $(L_1 + L_2)/\text{relative speed}$ · boat $(D + U)/2$, stream $(D - U)/2$
Mixture	Cheaper : dearer = $(d - m) : (m - c)$ · replacement $\times (1 - \frac{x}{C})^n$
Mensuration	Circle πr^2 , $2\pi r$ · triangle Heron · equilateral $(\sqrt{3}/4)a^2$ · cylinder $\pi r^2 h$ · cone $\frac{1}{3}\pi r^2 h$, $l = \sqrt{(r^2 + h^2)}$ · sphere $\frac{4}{3}\pi r^3$, $4\pi r^2$ · hemisphere total $3\pi r^2$
Algebra	$x + \frac{1}{x} = k \rightarrow x^2 + \frac{1}{x^2} = k^2 - 2$, $x^3 + \frac{1}{x^3} = k^3 - 3k$ · $a + b + c = 0 \rightarrow a^3 + b^3 + c^3 = 3abc$ · roots: sum $-b/a$, product c/a
Geometry	Exterior angle = sum of opposite interiors · $(n - 2)180$ · $360/n$ · diagonals $n(n - 3)/2$ · centre angle = $2 \times$ circumference angle · cyclic opposite 180°
Trigonometry	$\sin 30 = \frac{1}{2}$, $\sin 45 = \frac{1}{\sqrt{2}}$, $\sin 60 = \frac{\sqrt{3}}{2}$ · $\tan 30 = \frac{1}{\sqrt{3}}$, $\tan 60 = \sqrt{3}$ · $\sin^2 + \cos^2 = 1$ · $\sec^2 - \tan^2 = 1$ · $h = d \tan \theta$

Topic	Formulas
Statistics, probability	Median after sorting · mode = 3 median – 2 mean · P = favourable/total · 2 dice 36 · cards 52
Clock, calendar	$\theta = 30H - 5.5M $ · mirror time = 11:60 – time · odd days: year 1, leap 2, 100 yrs 5 · century leap only if $\div 400$

Reasoning rules

Topic	Rule
Series	Differences → second differences → ratios → squares/cubes → alternate terms
Alphabet	EJOTY = 5, 10, 15, 20, 25 · opposite letter: sum of positions = 27
Ranking	Total = left + right – 1 · from the other end = total – rank + 1
Direction	Right turn = clockwise · morning shadow points west · 3-4-5 triangles
Blood relations	Draw the family tree; mark gender (+/–) for every person
Seating	Facing north: their left = your left · circle facing centre: left = clockwise
Syllogism	A conclusion must hold in every diagram; 'possible' ≠ follows
Dice, cubes	Standard die opposite faces sum to 7 · painted n^3 cube: 8 corners, $12(n - 2)$, $6(n - 2)^2$, $(n - 2)^3$

GA static fact list (verified basics)

Area	Facts
Constitution	Adopted 26 Nov 1949, in force 26 Jan 1950 · Rajendra Prasad: President of the Constituent Assembly · B. R. Ambedkar: Drafting Committee chairman · 12 Schedules · 42nd Amendment (1976) added 'socialist', 'secular', 'integrity'
Articles	Fundamental Rights: Part III (Arts 12–35) · DPSPs: Part IV (36–51) · Fundamental Duties: Art 51A (now 11) · Art 17 abolishes untouchability · Art 21 life and liberty · Art 21A education · Art 32 constitutional remedies ('heart and soul') · emergencies: 352 national, 356 state, 360 financial
Parliament	Rajya Sabha maximum 250 (12 nominated) · President must be at least 35 · money bill: Art 110
Freedom movement	INC founded 1885 (A. O. Hume; first president W. C. Bonnerjee) · Partition of Bengal 1905 · Muslim League 1906 · Gandhi returned 1915 · Champaran 1917 · Jallianwala Bagh 13 Apr 1919 · Non-Cooperation 1920 · Chauri Chaura 1922 · Simon Commission 1928 · Purna Swaraj 1929 (Lahore) · Dandi March 12 Mar 1930 · Quit India 8 Aug 1942
Geography	Tropic of Cancer crosses 8 states · IST meridian 82.5° E (near Mirzapur) · largest state Rajasthan, smallest Goa · Wular largest freshwater lake · Chilika largest brackish lake · Indira Point southernmost · Tehri highest dam · Hirakud longest dam
Economy	RBI set up 1 Apr 1935, nationalised 1949, HQ Mumbai · First Five-Year Plan 1951 · NITI Aayog 1 Jan 2015 · GST 1 Jul 2017 · repo rate = rate at which RBI lends to banks
Railways	First passenger train 16 Apr 1853, Bori Bunder (Bombay) – Thane, 34 km · Railway Board 1905 · first Vande Bharat flagged off Feb 2019 (New Delhi – Varanasi) · National Rail Museum, New Delhi

Current affairs change every month and are not in this list — use your CA PDF headlines from the last 2 months.

Your mistake log — top 10 traps (write them here)

#	My trap	The rule that fixes it
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Exam-day checklist

When	Do
Tonight	Print the admit card · read its instructions (ID, photo, reporting time) · keep the original photo ID listed there · plan the route · sleep by 10:30 pm
Morning	20 min: formula sheet + your top-10 traps · light breakfast · leave early — reach before the reporting time
In the hall	Plan: GA → Science → Reasoning → Maths (or the order your mocks proved best) · two rounds · no blind guesses · ≈ 7 wrong is your whole budget
If a section feels hard	It is hard for everyone in that shift — normalisation adjusts for it. Move on; don't chase it.

Send me tonight

Nothing tonight. Sleep. After the exam, tell me how it went and which shift you had.